



Neuro Basic Profile; urine



Order: 200911-2008



Test: U200911-2008-1

Client #: 42497

Doctor: David Duizer, ND

Finlandia Health Centre

G104-2480 Spruce St

Vancouver, BC V6H 2P6 Canada

Patient: Jennifer Pei Rv Hao

Id: P202200204

Age: 26 **DOB:** 02/15/1994

Sex: Female

Sample Collection **Date/Time**

Date Collected 09/09/2020

Collection Period 2nd morning void

Date Received 09/11/2020

Date Reported 09/16/2020

Analyte	Result	Unit per Creatinine	L	WRI	H	Reference Interval
Serotonin	92.0	µg/g				60 – 125
Dopamine	184	µg/g				125 – 250
Norepinephrine	7.1	µg/g				22 – 50
Epinephrine	1.0	µg/g				1.6 – 8.3
Norepinephrine / Epinephrine ratio	7.1					< 13
Glutamate	18	µmol/g				12.0 – 45.0
Gamma-aminobutyrate (GABA)	3.3	µmol/g				2.0 – 5.6
Glycine	1398	µmol/g				450 – 2200
Histamine	15	µg/g				14 – 44
Phenethylamine (PEA)	10	nmol/g				32 – 84
Creatinine	73.2	mg/dL				30 – 225



Neurotransmitter Comments:

- Urinary neurotransmitter levels provide an overall assessment of the body's ability to make and break down neurotransmitters and are representative of whole body levels. Neurotransmitters are secreted all through the body, in neurons of both the central and peripheral nervous systems. The enzymes, cofactors and precursors in neurotransmitter metabolism in general are the same in the periphery and in the central nervous system. Therefore, alterations in urinary neurotransmitter levels assessed in urine provide important clinical information, and may be associated with many symptoms including cognitive and mood concerns, diminished drive, fatigue and sleep difficulties, cravings, addictions and pain.
- Low norepinephrine and low epinephrine may be associated with depression and mood changes as well as fatigue, difficulty concentrating, decreased ability to stay focused on tasks and diminished sense of personal/professional drive. Norepinephrine is converted from dopamine requiring vitamin C, copper and niacin (B3). L-tyrosine, L-theanine and Mucuna pruriens influence this pathway.
- Low phenethylamine (PEA) may be associated with depression, attention deficits and hyperactivity (ADHD), Parkinson's disease and bipolar disorder. Phenylalanine is the precursor amino acid to PEA, and vitamin B6 is a required co-factor in the conversion to this primary trace amine. Use of Reserpine can result in depletion of PEA.
- Considerations to address the demonstrated imbalances beyond the identified co-factors and amino acid precursors may include dosage adjustments if indicated, as well as nervine and adaptogenic herbs, methylation support, vitamin D, and gastrointestinal health optimization.

Notes:

Results are creatinine corrected to account for urine dilution variations. Creatinine is not meant to be used as an indicator of renal function.

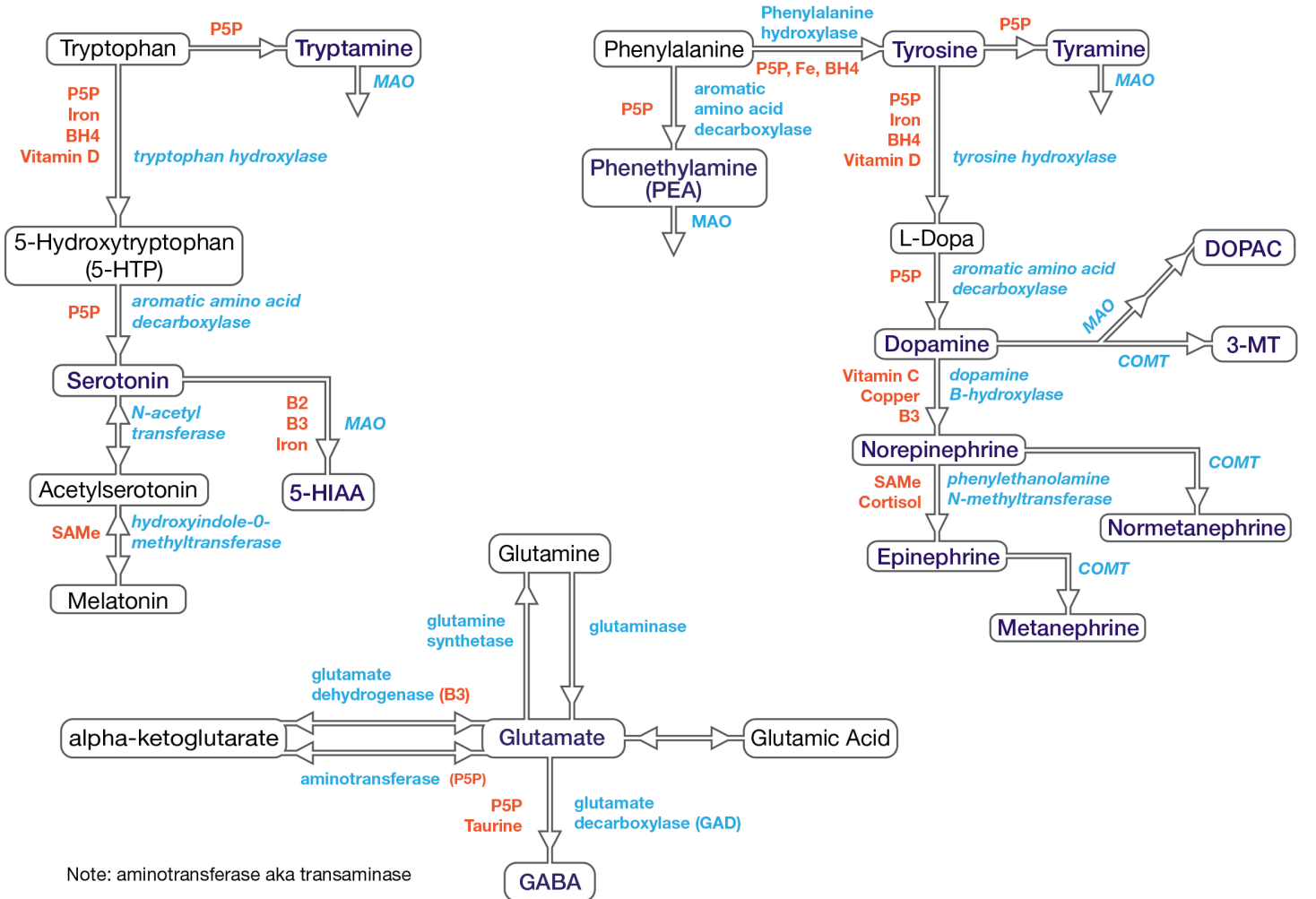
RI= Reference Interval, L (blue)= Low (below RI), WRI (green)= Within RI (optimal), WRI (yellow)= Within RI (not optimal), H (red)= High (above RI)

Methodology: LCMS QQQ, Creatinine by Jaffe Reaction

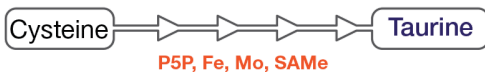
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NT Neurotransmitter Pathways



"glycine cleavage system"



KEY

MAO = monoamine oxidase

Cofactors for MAO: **B2, B3, P5P, Fe, Mg**

COMT = catechol-o-methyl-transferase

Cofactors for COMT: **SAmE, Mg**

P5P = (pyridoxal-5-phosphate) activated form of vitamin B6

BH4 = (tetrahydrobiopterin)

Endogenous levels can be supported with SAmE, vitamin B3, C, Mo, Zn

MTHF = (methyltetrahydrofolate) active form of folate.

SAmE = endogenous levels can be supported with Mg, MTHF, and methylcobalamin supplementation.

Cofactors = ■

Enzymes = ■